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Vehicle detection and tracking based on improved YOLOv8

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ABSTRACT With the rapid increase in transportation pressure, the demand for efficient traffic recognition and tracking systems is growing. Traditional methods have certain limitations when dealing with complex situations in traffic scenes, such as large weights and insufficient detection accuracies, *et al.* Therefore, we propose a novel method based on YOLOv8n. Firstly we introduced SCC_Detect based on SCConv on the detection head to reduce the computation of redundant features. Then we replaced the convolutional kernel with a dual convolutional kernel to construct a lightweight deep neural network. Subsequently, the Focaler-ElIoU loss function is introduced to improve the accuracy. The BotSORT tracker is embedded in the period of inference, which achieves more accurate and stable recognition and tracking results in the traffic scene. The experimental results show that the proposed model reduces parameters and weight by approximately 36.5% and 25% respectively at the expense of only 0.2% mAP@0.5 compared with YOLOv8n on the UA-DETRAC dataset. In terms of tracking, MOTA, IDF1 and MOTP of the BoTSORT algorithm on the test video were superior to those of DeepSORT and StrongSORT. The accuracy was improved, and the number of lost tracks was reduced. It has a high practical value and application prospects in traffic detection and deployment.

INDEX TERMS Vehicle Tracking, Multi-Object Track, Lightweight model, YOLOv8, Vehicle Detection, BotSORT

I. INTRODUCTION

The total number of civilian vehicles in China reached 336.18 million, according to the statistics from the National Bureau of Statistics of China by the end of 2023, an increase of 17.14 million from the previous year. With the continuous advancement of urbanization and sharp increase in the number of vehicles, traffic management and safety issues have increasingly become a focus of public concern.

Traffic flow detection, which involves the real-time monitoring and analysis of vehicle flow on roads, is a core component of modern Intelligent Transportation Systems (ITS)[1]. This is an important tool in traffic management. By monitoring traffic conditions in real-time, traffic management departments can obtain the traffic volume at a specific road section and time, which facilitates supervision and statistics. It also allows for quick response and effective handling of traffic congestion and accidents. Navigation systems that incorporate traffic flow data can deliver up-to-the-minute updates on road conditions to drivers, helping them to avoid congested routes and choose the best driving paths, thereby saving time and reducing fuel consumption.

Vehicle tracking, which involves continuous monitoring and recording of vehicle movement on roads, is also a key function of the ITS. Traffic flow tracking contributes to improving road safety by monitoring vehicle driving routes and behaviors in real-time. This helps to identify potential traffic accident risks, such as improper lane changes and running red lights. Traffic flow tracking is crucial for enabling communication between vehicles and between vehicles and infrastructure, which helps improve traffic efficiency, reduce accidents, and realize advanced functions such as autonomous driving. The deployment of traffic flow tracking and detection has been facilitated by the widespread installation of surveillance cameras on most road sections. The importance of traffic flow detection is self-evident. They play a crucial role in improving traffic efficiency, optimizing resource allocation, and ensuring public safety at multiple levels. This significantly contributes to building a more intelligent, efficient, and safe traffic environment.

Currently, in traffic scenarios, the primary method mostly relies on surveillance cameras installed on the roads. The models for vehicle detection are very complex, requiring high

In terms of detection, where the relative position of the camera in road traffic is essentially unchanged, and there is only vehicle movement in areas such as highways and national roads, traffic flow detection is relatively easier. Reference[11] replaced the Spatial Pyramid Pooling (SPP) module with the Atrous Spatial Pyramid Pooling (ASPP)[12] module in the YOLOv5s algorithm, thereby increasing the receptive field of different-sized images to extract multi-scale context information. They also added a detection layer to learn traffic flow features. Lin et al.[13] adopted Depthwise Separable Convolution based on YOLOv5s, reducing the model's computational load and weight size without sacrificing accuracy.

In object detection, early detection methods rely on manually designed feature extraction, which does not achieve high detection accuracy. The AlexNet[7] successfully promoted the development of Convolutional Neural Networks[8]. Reference [9] introduced an R-CNN algorithm based on CNN,



Zhao et al.[14] based on YOLOv5, replaced the bottleneck CSP structure with the Ghostnet[15] structure, introduced a coordinate attention mechanism, and also employed Distance-IoU non-maximum suppression to enhance the network's detection capability and target feature extraction ability.

Traffic conditions are relatively complex with respect to traffic tracking, owing to weather, traffic jams, pedestrians, and other factors that can cause varying degrees of occlusion. Reference[16] completed multi-target tracking in two stages. The initial processing employed YOLOX[17] for object detection, with the detected data fed into the DeepSORT[18] algorithm in the subsequent stage for the Kalman filter(KF) input. Subsequently, a deep convolutional neural network was utilized to extract features from the detected frames, which were then matched against the stored features to facilitate video analysis. Reference[19] combined YOLOv7 with ByteTrack[20] and designed a backbone network based on RepVGG[21] called EfficientRep[22] and RepPAN. Experiments proved that ByteTrack significantly improved the detection rate compared to DeepSORT.

III. Our method

Owing to the differences in the shape, size, and other attributes of complex road environment traffic, as well as the problems of a large model size, a method based on the improved YOLOv8n is proposed. Because embedded devices are limited by memory size and computational resources, their deployment of convolutional neural networks is constrained. We first introduced SCC_Detect based on SCConv[23] instead of the original detect head to simultaneously enhance the ability of feature extraction and reduce parameters to increase the simplicity of deployment. Then we replace the convolutional kernel with a double convolutional kernel, which makes the backbone feature network lightweight while

reducing the loss of accuracy. For the loss function, we selected Focaler-Eiou[24], which improves the performance of the detector by focusing on different regression samples. The entire network structure of the proposed YOLOv8 model is illustrated in Fig. 1. Combining the special needs of traffic scenarios with the cutting-edge BOTSORT[25] algorithm, we optimize the tracking and detection capability of dynamic targets and the accuracy of traffic tracking under multiple targets. At the same time, it is applicable to a variety of complex traffic environments, such as urban roads, highways, and intersections.

A. SCC-Detect

To improve the head of YOLOv8, we designed a detection head based on SCConv. SCConv comprises two units: Spatial Reconstruction Unit (SRU) and Channel Reconstruction Unit (CRU). The architecture of SCConv is shown in Fig. 2. Feature X at the intermediate stage in the bottleneck residual block is first operated by SRU to acquire the enhanced spatial detail characteristic X^w , and then CRU is utilized to derive the refined channel feature Y. Exploiting the spatial and channel redundancy inherent in the SCConv module's feature set, it can be smoothly incorporated into any network based on the CNN. The purpose of this fusion is to reduce the overlap in intermediate feature maps and to improve the feature detection proficiency of the CNN.

SRU with a separation-reconfiguration operation is used for space redundancy. This division is intended to differentiate between feature maps that are dense in informative content and those that contain minimal spatial-related information. The formulas for the separation operation are shown in (1) and (2).

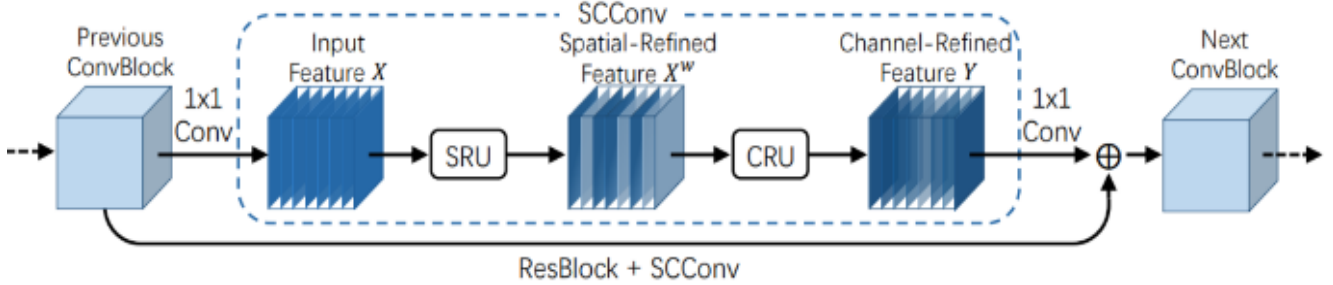


FIGURE 2. The network architecture of SCConv.

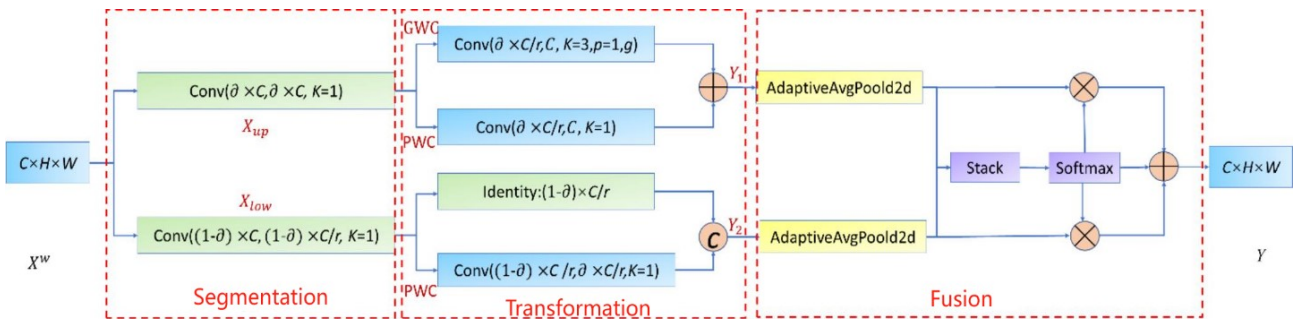


FIGURE 3. CRU implementation diagram.

$$X_{out} = GN(X) = \gamma \frac{X - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta \quad (1)$$

$$W_\gamma = W_i = \frac{\gamma_i}{\sum_j \gamma_j}, i, j = 1, 2, \dots, C \quad (2)$$

Where GN represents the group normalization layer. μ and σ denote the average and the standard deviation of X , respectively. ϵ is a small positive number added to stabilize the division, γ and β are trainable affine transformations. We first normalized the input features, and then the learnable parameter γ was applied to assess the spatial pixel values for each batch and channel, normalized with the corresponding weights W_γ . The value of W_γ multiplied by X_{out} was mapped to (0, 1). A threshold value was used to gate the values. W_γ greater than the threshold is considered a useful feature set for W_1 , and W_γ less than the threshold is considered a useless feature set for W_2 . The weights W_1 and W_2 can be obtained using (3).

$$W = GateSigmoidW_\gamma GN(X) \quad (3)$$

Finally, (4)-(8) use cross-reconstruction to enhance the useful features in the spatial dimension to suppress redundant features and obtain spatially refined features X^w .

$$X_1^w = W_1 \otimes X \quad (4)$$

$$X_2^w = W_2 \otimes X \quad (5)$$

$$X_{11}^w \oplus X_{22}^w = X^{w1} \quad (6)$$

$$X_{21}^w \oplus X_{12}^w = X^{w2} \quad (7)$$

$$X^{w1} \cup X^{w2} = X^w \quad (8)$$

Where $\otimes$ denotes element-by-element multiplication, $\oplus$ denotes element-by-element addition, and $\cup$ denotes Concat module.

After the processing of the SRU, although the features with abundant information are distinguished from those with lesser information, the resulting refined features X^w are still redundant in the channel, and the CRU is used for the segmentation-transformation-fusion strategy to reduce channel redundancy. The implementation of CRU is shown in Fig. 3.

The features were further classified into rich feature extractor X_{up} and shallow feature extractor X_{low} using segmentation. X_{up} uses the efficient convolution operation group-wise convolution (GWC) combined with point-wise convolution (PWC) instead of the standard expensive $K \times K$ convolution,

whereas X_{low} employs the cost-effective 1×1 PWC operation to produce feature maps with superficial hidden details to obtain Y_2 . Subsequently, Y_1 and Y_2 were fused to obtain Y with more valid information. Therefore, input feature X can reduce information redundancy and extract valid information after SRU and CRU.

In the improved YOLOv8 model, SCConv is designed as a pluggable module, which allows the model to introduce SCConv while maintaining the original structure, and is able to reduce the parameters of the model while improving detection accuracy, which is especially important for resource-constrained devices.

B. Backbone and neck structures

In our proposed method, we employed a new dual convolutional filter architecture to replace the traditional convolutional operation and improve the efficiency of the backbone network. This approach, called DualConv[26], combines 3×3 and 1×1 convolution kernels, with a 3×3 convolution that provides spatial information processing. The 1×1 convolution kernel is used to integrate these features and reduce the parameters of the model. DualConv reduces the complexity of the model by employing a group convolution technique, where the input and output feature maps are divided

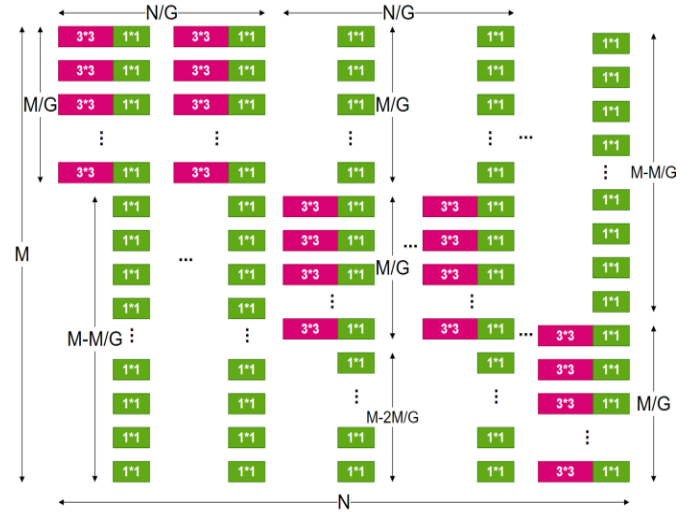


FIGURE 4. Illustration of Dual convolution.

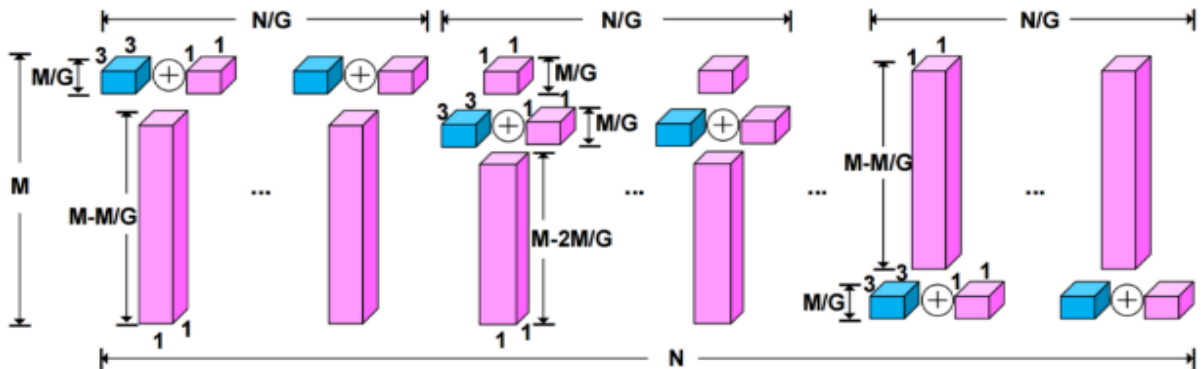


FIGURE 5. Kernels' configuration of Dual convolution.

into multiple groups, and the convolution filter for each group processes only a portion of the corresponding input feature map. This reduces the model's complexity and facilitates the effective arrangement of filters.

As shown in Fig. 4, the channels (M) in the input feature map signifies its depth, while the convolutional kernels and output channels (N) define the depth of the resulting feature map. In addition, the groups (G) in the dual convolution determine how the filters are grouped. Each group handles the entire input feature map processing, where 3×3 and 1×1 convolution kernels process the M/G input feature map channels in parallel. The left channels (M-M/G) were filtered through distinct 1×1 convolutional kernels. In this manner, the results of the 3×3 and 1×1 convolutional kernels were summed to generate the final output, as shown by the $\oplus$ symbols in Fig. 5. Each group's convolutional kernels handle a segment of the input channels independently, and the outputs are then merged, resulting in efficient flow and integration of information across different feature map channels. This architectural design preserves not only the network's depth and feature representation ability, but also reduces the parameters of the original backbone network model, making it suitable for resource-constrained environments.

C. Regression Loss

In YOLOv8, CIOU[27] is introduced as a loss function. It considers the differences between the centroid, width and height of the detection frame into account in order to accurately reflect the shape of the object. However, in a traffic flow detection scenario, the pixel ratios of the background and the target vehicle are very different from each other, which leads to an imbalance of positive and negative samples, and can easily result in the model overfitting a large number of simple background samples while neglecting the sparse detection capability of the target vehicles. The Focaler-ElIoU loss can effectively alleviate the sample imbalance problem and improve the model's detection accuracy of the target vehicle by introducing a method of dynamically adjusting the weights. First, ElIoU[28] redefines shape loss based on CIOU which is defined in (9).

$$ElIoU = IoU - \frac{\rho^2(b, b^{gt})}{c^2} - \frac{\rho^2(w, w^{gt})}{(w^c)^2} - \frac{\rho^2(h, h^{gt})}{(h^c)^2} \quad (9)$$

Where w^c and h^c represent the dimensions of the smallest enclosing box that encompasses both the ground truth and anchor boxes, respectively. The IoU is defined in (10).

$$IoU = \frac{|B \cap B^{gt}|}{|B \cup B^{gt}|} \quad (10)$$

Where B represents the predicted box and B^{gt} represent the GT box. To improve the edge regression, we need to reconstruct the IoU loss, as shown in (11).

$$IoU^{focaler} = \begin{cases} 0, & IoU < d \\ \frac{IoU-d}{u-d}, & d \ll IoU \ll u \\ 1, & IoU > u \end{cases} \quad (11)$$

Where $[d, u] \in [0, 1]$. The $L_{Focaler-IoU}$ is defined in (12).

$$L_{Focaler-IoU} = 1 - IoU^{focaler} \quad (12)$$

$$L_{Focaler-ElIoU} = L_{ElIoU} + IoU - IoU^{focaler} \quad (13)$$

$L_{Focaler-ElIoU}$, which applies Focaler-IoU to the current IoU-based bounding box regression loss function is defined in (13).

IV. Traffic Tracking Method

The multi-object tracking system monitors objects in a scene with a detection algorithm that predicts the paths of targets in a video and assigns a distinct ID to each target. Motion model and appearance estimation are two steps mainly using in tracking. The bounding boxes for the motion trajectory in following frames were forecasted with the KF. The target localization and appearance model, by predicting the Intersection over Union (IoU) between the track bounding boxes and detection bounding boxes, as well as reidentification (ReID), associates the latest frame detection results of the video sequence with the current set of trajectories. In our study, we selected BoTSORT, an improved version of ByteTrack. It adopts the overall SORT framework and combines the benefits of incorporating the motion characteristics and appearance information. By adding a KF state vector and camera motion compensation, stable tracking of multiple vehicles in complex scenes is achieved.

To simulate motion trajectories in images, a discrete KF with a constant velocity model is typically employed to model the motion of moving detection targets. The KF state vector of SORT is a 7-tuple, whereas the KF state vector of DeepSORT is an 8-tuple, which changes the prediction of the aspect ratio in the fourth dimension of the SORT tuple to the prediction of the height of the bounding box. To achieve a better performance, BoTSORT redefines the state vector of the KF. BoTSORT changes the aspect ratio to predict the width, and directly estimates the width and height of the bounding box.

A. Camera motion compensation(CMC)

The tracker relies on the overlap between the predicted target motion trajectory bounding box and detected target trajectory bounding box. A moving camera causes video image instability, resulting in alterations in the bounding box's position and leading to inaccurate prediction and matching. Outdoor cameras are easily affected by wind and rain, which can cause a slight shaking. Consequently, the pixel position of the bounding box may change, leading to a loss in vehicle tracking. Therefore, BoTSORT utilizes Global Motion Compensation (GMC). This technology originates from the OpenCV visual library and is a video stabilization module that includes an affine transformation algorithm. However, affine transformation can only affect the center position of the bounding box, and the remaining parts require correction using a camera motion rectification equation. The related Equations are as follows.

$$K_k = P'_{k|k-1} H_k^T (H_k P'_{k|k-1} H_k^T + R_k)^{-1} \quad (14)$$

$$\hat{x}_{k|k} = \hat{x}'_{k|k-1} + K_k (z_k - H_k \hat{x}'_{k|k-1}) \quad (15)$$

$$P_{k|k} = (I - K_k H_k P'_{k|k-1}) \quad (16)$$

Where z_k is the measurement vector, I represents the identity matrix, R_k denotes the covariance matrix of the measurement error, H_k represents the matrix that translates the state vector into the measurement domain for refining the state prediction in the measurement update phase of the KF, while K_k is the matrix that blends the anticipated state estimate with the observed measurement results. $\hat{x}_{k|k-1}, \hat{x}'_{k|k-1}$ are the KF predicted state vector before and after camera motion compensation at time step k . $P_{k|k-1}, P'_{k|k-1}$ are the predicted covariance matrix of KF before and after correction.

B. IoU+ReID Fusion

In vehicle flow detection, BoTSORT integrates the appearance of vehicles with the tracker. Owing to the high likelihood of occlusion among vehicles, low-confidence detections were discarded. BoTSORT uses the IoU+ReID fusion mechanism to calculate the cosine similarity. For the IoU scores, only a high cosine similarity was accepted. The formulas for the IoU-ReID fusion pipeline are defined in (17) and (18).

$$C_{i,j} = \min(d_{i,j}^{iou}, \hat{d}_{i,j}^{cos}) \quad (17)$$

$$\hat{d}_{i,j}^{cos} = \begin{cases} \frac{1}{2} d_{i,j}^{cos}, & d_{i,j}^{cos} < \theta_{emb} \text{ and } d_{i,j}^{iou} < \theta_{iou} \\ 1, & \text{others} \end{cases} \quad (18)$$

Where $C_{i,j}$ is the (i, j) element of the cost matrix C and $\hat{d}_{i,j}^{cos}$ is the new appearance cost. $d_{i,j}^{iou}$ represents the IoU distance, which is the measure of overlap between the i -th predicted track bounding box and the j -th detected bounding box, signifying the motion cost.

C. Tracking Algorithm Process

The BoTSORT tracking algorithm process is as follows: Before tracking begins, BoTSORT initializes all targets, using a pretrained object detection model to detect all targets in each frame. By setting a low confidence threshold, the outcomes of detection are categorized into high-confidence and low-confidence detections. During tracking, BoTSORT combines object detection and motion prediction for multi-object tracking. First, the KF is used for the state prediction of each existing track to estimate the position of the target in the current frame. Next, the Hungarian algorithm was used to associate the predicted tracks with the detections in the current frame, typically based on the spatial position and appearance features of the target. Data association was mainly conducted between high-confidence detections and predicted tracks, and the remaining low-confidence detections were associated with unmatched tracks. For unmatched high-confidence detections, new target tracks are created; for tracks that are unmatched for consecutive frames, they are marked as lost and deleted after exceeding a certain time. Finally, post-processing was performed to obtain the tracking results.

V. Experiments

A. UA-DETRAC

The UA-DETRAC[29] dataset is an extensive dataset utilized for identifying and tracking vehicles. The dataset was mainly filmed on road overpasses in Beijing and Tianjin (Jingjinji region scenes) and manually annotated with 8250 vehicles and more than 1.21 million object bounding boxes, totaling 82,050 images. It contains over 14,000 vehicle instances and provides precise annotations for vehicle detection and tracking. The dataset includes several weather conditions such as night, clear, and rainy. The vehicle images in the dataset have three levels of occlusion: completely visible, partially occluded by other vehicles, and partially occluded by the scene. Vehicles are categorized into four classes: cars, buses, trucks, and others. Each annotated image has an associated XML format annotation file, which contains information regarding the category of vehicles within the image and the location of the annotation boxes. We changed the XML format annotations to TXT format annotations. The split ratio between the training and validation sets was 9:1. UA-DETRAC already had a test set. Fig. 6 shows an example of images from the dataset.

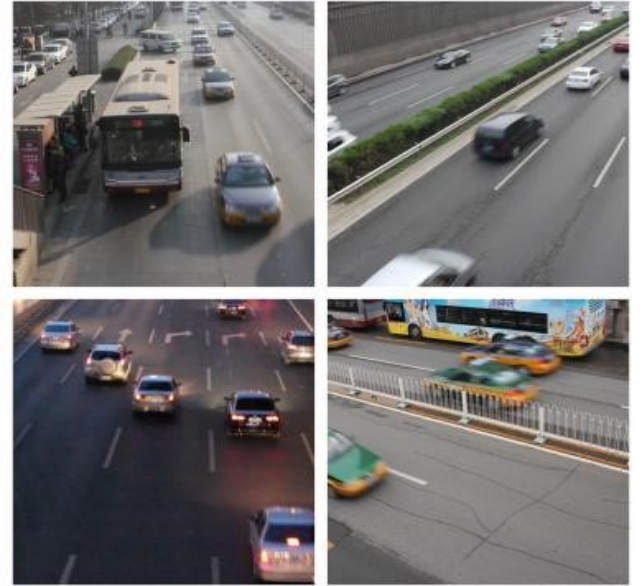


FIGURE 6. An example of part of UA-DETRAC

B. Experiment Platform and Evaluation Metrics

The experimental computer processor used was an Intel® Core™ i5-13600KF CPU @ 3.50 GHz, the GPU was an NVIDIA RTX 4070 Ti 12GB, the operating system was Windows 11, the deep learning framework was PyTorch 2.2.1, the Python version was 3.9.6, and the CUDA version was 12.1. The settings of the YOLOv8 model training parameters were as follows: batch size was set to 32, number of training epoches was 200, image size was set to default 640, initial learning rate was 0.01, weight decay coefficient was 0.0005,

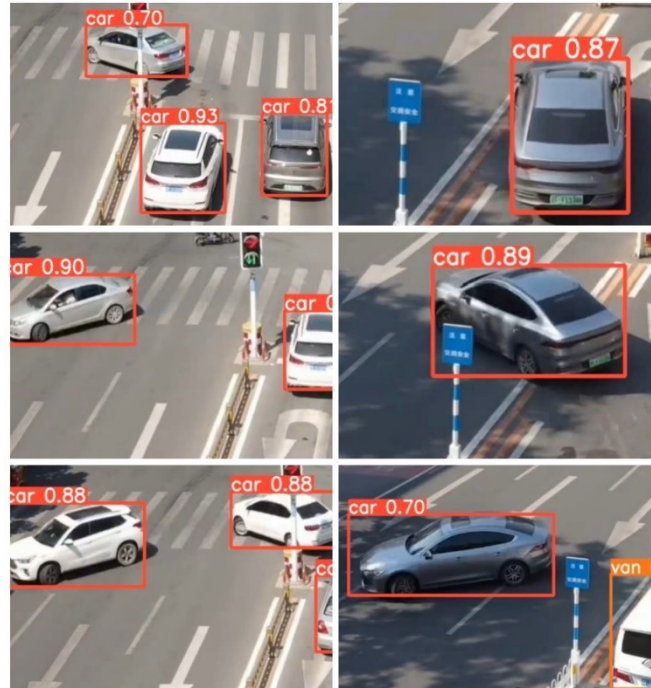


FIGURE 7. Detection examples of vehicles at various angles to the camera.



FIGURE 8. The detection results inferred by YOLOv8 and the proposed model.

momentum was 0.937, and SGD was used as the optimizer for iterative training.

For the BoTSORT parameters, the track high thresh (threshold for the first association) was set to 0.8, the track low thresh (threshold for the second association) was set to 0.5, and the

new track thresh (threshold for initiating a new track if the detection does not match any tracks) was set to 0.6. The main evaluation metrics used to verify the improved vehicle target detection performance of the model were Precision (P), Recall (R), mean Average Precision (mAP), parameters, and weight. The mAP is calculated with an IoU threshold of 0.5 (mAP@0.5) and at different IoU thresholds ranging from 0.5 to 0.95 (mAP@0.5-0.95) to further evaluate the model's precision. For vehicle tracking, we selected a few sequences of consecutive frame images from the test set to form a video and conducted MOT testing on this video. The selected evaluation metrics were False Positive (FP), False Negative (FN), Multiple Object Tracking Accuracy (MOTA), Multiple Object Tracking Precision (MOTP), Identification F1 score (IDF1), and Mostly Lost Ratio (MLR). Where P, R, and mAP are higher, indicating that the target detection and recognition are more accurate. Higher values for MOTA, MOTP, and IDF1, along with lower FP, FN and MLR, reflect better tracking performance.

C. Experimental Results

We first compared the improved model with the original model, as shown in Table 1. Fig. 7 shows the detection effect of the steering vehicles detected by the proposed model. Fig. 8 shows the inference of the YOLOv8 model and the model we proposed on the test set. We can see that our proposed model has relatively fewer missed detections compared to YOLOv8, while the confidence level is slightly lower than that of YOLOv8. The experiment compared P, R, and mAP values. We also compared the model weights to determine ease of deployment. It can be seen that our proposed model achieved mAP@0.5 and mAP@0.5-0.95 of 99.0% and 90.1%, respectively, which is a decrease of 0.2% and 0.5% compared to the YOLOv8n model. However, the parameters decreased by 3.7GFLOPs, and the weight was only 4.5MB.

TABLE 1
PROPOSED MODEL COMPARED TO YOLOV8

Evaluation Metrics	YOLOv8	Proposed
P(%)	97.8	96.7
R(%)	97.4	97.5
mAP@0.5(%)	99.2	99.0
mAP@0.5-0.95(%)	90.7	90.2
GFLOPs(G)	8.2	5.2
Weight(MB)	6	4.5

TABLE 2
DIFFERENT MODELS EXPERIMENT RESULTS.

Model	P(%)	R(%)	mAP@0.5(%)	mAP@0.5-0.95(%)	GFLOPs(G)	Weight (MB)
Faster-RCNN	-	-	90.3	-	201.1	309
YOLOv3	92.1	91.9	93.41	81.2	141	124.4
YOLOv5s	98.1	98.3	99.3	86.5	16.4	14.4
YOLOv5s-ADD-ASPP[11]	93.7	94.2	93.9	-	-	-
YOLOv5s_DSC[13]	98.0	97.6	99.3	85.1	12.3	11.5
YOLOv5s_GhostNet_DIoU-NMS_CA[14]	96.6	95.1	98.7	84.3	11.6	10.8
YOLOv7-tiny	97.0	97.1	99.0	88.2	13.2	12.3
YOLOv8n	97.8	97.4	99.2	90.7	8.2	6
Proposed	96.7	97.5	99.0	90.2	5.2	4.5

D. Comparative Experiment

We trained YOLOv3, YOLOv5s, YOLOv7-tiny, YOLOv8n, and the proposed model on UA-DETRAC, as well as the methods proposed in other studies using the same dataset. The experimental results are presented in Table 2. Compared to Faster R-CNN and YOLOv3, our model achieved significant advantages in terms of both accuracy and parameter quantity. For YOLOv5s and YOLOv7, although there is not much difference in accuracy, the proposed model has made some progress in complexity and weight size. Compared with YOLOv5s_DSC, we decreased the parameters and weight by 4.1GFLOPs and 7MB, respectively, whereas mAP@0.5-0.95% increased by 5.1%. Although the mAP@0.5 of the model proposed in [13] is slightly higher, its GFLOPs and weight size are 7.1G and 7MB higher than those of the proposed model, respectively. The method in [14] also decreased the parameters, however, we still decreased the parameters by 6.4 GFLOPs.

E. Ablation experiments of each improved module.

To assess the efficacy of the proposed model, we designed comparative experiments using the improved methods DualConv, SCC_Detect, and Focaler_EIoU, as shown in Table 3. The combination of YOLOv8n and DualConv shows a 1.5% improvement in mAP@0.5-0.95% for vehicle detection compared with YOLOv8n, indicating that dual convolution can better integrate features for vehicles in different environments. Adding the SCC_Detect head to YOLOv8n does not result in a significant improvement in accuracy, however, the computational complexity of the model and weight are significantly reduced, proving that SCConv can reduce redundancy while maintaining accuracy. When using Focaler_EIoU, mAP@0.5-0.95% increased by 1.7%, indicating that the loss function we selected played a role in improving accuracy. Moreover, as shown in Fig. 9, Focaler_EIoU converges faster, and the object bounding box loss, distributional feature loss, and classification loss are more effective.

F. Tracking experiments

TABLE 3
COMPARISON EXPERIMENT ON THE ROLE OF EACH PART IN THE PROPOSED MODEL.

Model	P(%)	R(%)	mAP@0.5(%)	mAP@0.5-0.95(%)	GFLOPS(G)	Weight (MB)
YOLOv8n	97.8	97.4	99.2	90.8	8.2	6
YOLOv8n + DualConv	97.0	98.1	99.2	92.3	7.6	5.5
YOLOv8n + SCC_Detect	98.3	97.8	99.0	91.3	5.8	5
YOLOv8n + Focaler_EIoU	98.0	98.0	99.3	92.5	8.2	6.2
Proposed	96.7	97.5	99.0	90.2	5.2	4.5

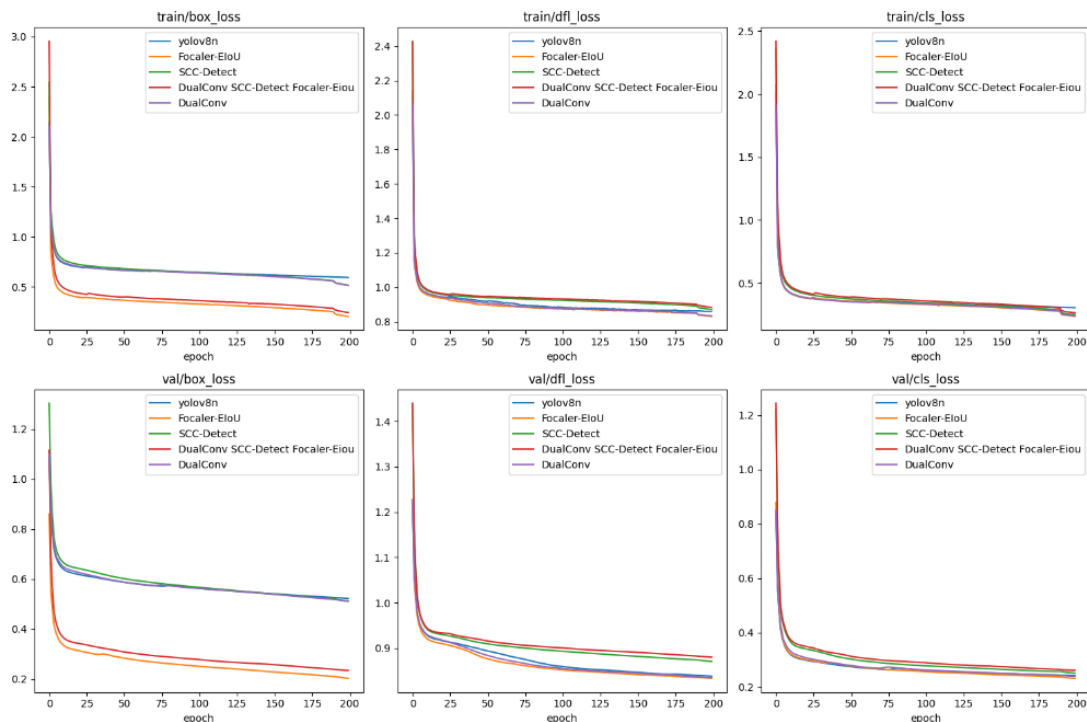


FIGURE 9. The performance of Focaler_EIoU loss function.

TABLE 4
EXPERIMENT RESULTS OF DIFFERENT ALGORITHMS.

Video Number	Model	FP	FN	ID Switch	IDF1(%)	MOTA(%)	MOTP(%)	MLR(%)
1	DeepSORT	1924	688	27	87.7	88.1	91.2	0.2
	ByteTrack	1884	636	21	90.0	89.9	91.7	0.1
	BoTSORT	1868	437	18	91.4	90.7	90.8	0.1
2	DeepSORT	24	24746	34	40.6	38.7	88.1	5.2
	ByteTrack	21	21839	27	42.4	40.7	89.9	4.3
	BoTSORT	2	19506	22	45.6	46.9	92.3	3.6
3	DeepSORT	1856	177	12	60.1	66.5	90.4	0
	ByteTrack	2797	183	11	63.7	51.2	93.6	0
	BoTSORT	1409	16	6	64.0	72.3	92.9	0

For the traffic flow tracking experiment, we selected the advanced DeepSORT, ByteTrack, for comparison with the BoTSORT. To test the performance and robustness of the BoTSORT algorithm, we inferred real-world scenarios within the test set and selected the same frame from the inference results. The inference results are shown in Fig. 10. By comparing the circled areas, we can see that owing to occlusions caused by vehicles at the camera's angle of view, which increases the difficulty of recognition, both DeepSORT

and ByteTrack have some missed detections. However, BoTSORT does not have any missed detections, which benefits from data association and trajectory management. In terms of the evaluation metrics, we selected some full segment of the traffic flow videos for comparison. The results are presented in Table 4. We can conclude that BoTSORT has a slightly lower MOTP than ByteTrack in videos 1 and 3, but it has fewer false negatives and false positives than ByteTrack, and its IDF1, MOTA, and other metrics are superior to both



FIGURE 10. Tracking example of DeepSORT, ByteTrack and BoTSORT.

ByteTrack and DeepSORT. In the detection of video 2, all three algorithms have a higher number of false negatives, which may be due to the fact that the time is at night and there are more vehicles, leading to congestion, which increases the difficulty of tracking and somewhat reduces the tracking accuracy. However, BoTSORT still has advantages in all metrics, owing to its excellent trajectory management. Overall, when uncertain factors arise in the environment, the BoTSORT algorithm can achieve excellent results among the tracking algorithms, mainly due to its CMC module, which reduces the prediction deviation of vehicle tracking under camera shake conditions, improves the accuracy of outdoor vehicle tracking, and maintains a high level of recognition and tracking accuracy in crowded scenes. In summary, the tracking method proposed in this study was generally superior to the ByteTrack and StrongSORT algorithms.

VI. Conclusion

This study introduced a new method for improving vehicle flow detection and tracking based on YOLOv8. The proposed method is primarily aimed at addressing issues such as low accuracy in traffic monitoring, high computational load of the model, and large weight. We transformed SCConv into the

detection head, then replaced the convolutional kernels with DualConv, and finally used Focaler_EIoU to improve the accuracy. We achieved high precision on the dataset while significantly reducing the computational complexity. The BoTSORT algorithm was utilized to achieve precise tracking of traffic flow, taking the behavior category information detected by the improved YOLOv8n as input into the BoTSORT algorithm, enabling multi-object recognition and tracking of traffic flow in complex outdoor environments. The test outcomes show that our proposed model exhibits strong efficacy in accuracy and ease of deployment. The tracking method we introduced yields excellent results and can be deployed in many fields, not just traffic monitoring. Above all, we are still looking for superior algorithms to improve traffic monitoring.

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